

Hessian Structure of a Linear Bigram Model

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Abstract

This note derives the gradient and Hessian of an untied two-layer linear bigram model trained with the full-vector mean squared error. The diagonal Hessian blocks reveal two distinct effects of data imbalance. Across embedding columns, block heterogeneity is determined solely by the input-token distribution. Across rows of the language-model head, all Hessian blocks are identical for arbitrary input and output distributions: the input distribution changes their common value, whereas the conditional target distribution does not enter these blocks. The conditional target distribution affects the gradient and the residual-dependent part of the cross-layer Hessian.

1 Problem Setting

We study population risk minimization for bigram prediction. We first specify the data-generating distribution, which determines the form of token imbalance, and then introduce the two-layer linear network and its training objective.

Bigram prediction problem and data distribution. Let C be the number of token classes and let $\mathbf{e}_1, \dots, \mathbf{e}_C \in \mathbb{R}^C$ be the standard basis vectors. A bigram sample consists of an input token $x = \mathbf{e}_i$ and its next-token target $y = \mathbf{e}_j$. We assume that (x, y) follows a population distribution $\mathcal{D}$ specified by the input-token probabilities $\pi = (\pi_1, \dots, \pi_C)$ and the conditional next-token probabilities

$$\pi_i = \Pr(x = \mathbf{e}_i), \quad \pi_{j|i} = \Pr(y = \mathbf{e}_j \mid x = \mathbf{e}_i),$$

where $\sum_i \pi_i = 1$ and $\sum_j \pi_{j|i} = 1$ for every input class with $\pi_i > 0$. The joint and output-token probabilities are therefore

$$\Pr(x = \mathbf{e}_i, y = \mathbf{e}_j) = \pi_i \pi_{j|i}, \quad \pi_j^{\text{out}} = \Pr(y = \mathbf{e}_j) = \sum_{i=1}^C \pi_i \pi_{j|i}.$$

For each input class, collect its conditional next-token distribution in

$$\boldsymbol{\pi}_{\cdot|i} = [\pi_{1|i} \quad \cdots \quad \pi_{C|i}]^\top = \mathbb{E}[y \mid x = \mathbf{e}_i].$$

We also define the diagonal matrix of input-token probabilities as

$$D_\pi = \text{diag}(\pi_1, \dots, \pi_C).$$

In this formulation, input-token imbalance corresponds to nonuniform π_i , while the conditional probabilities $\pi_{j|i}$ determine how the next-token distribution varies with the input.

Two-layer linear network and training objective. Let d be the hidden dimension. The network consists of an embedding matrix W and a language-model head V :

$$W \in \mathbb{R}^{d \times C}, \quad V \in \mathbb{R}^{C \times d}.$$

For an input $x = \mathbf{e}_i$, the i -th embedding column $w_i = W\mathbf{e}_i$ is the hidden representation, and the network prediction is

$$h = Wx = w_i, \quad f(V, W; x) = Vh = VWx.$$

Given a target y , define the prediction error by

$$e = f(V, W; x) - y = VWx - y.$$

Here e denotes the prediction error, whereas the bold symbol $\mathbf{e}_i$ denotes a standard basis vector.

We train both layers by minimizing the population full-vector MSE:

$$\mathcal{L}(V, W) = \frac{1}{2} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\|VWx - y\|_2^2] = \frac{1}{2} \mathbb{E}_{(x,y) \sim \mathcal{D}} [e^\top e]. \quad (1)$$

Conditioning (1) on the input token yields

$$\mathcal{L}(V, W) = \frac{1}{2} \sum_{i=1}^C \pi_i \|Vw_i - \boldsymbol{\pi}_{\cdot|i}\|_2^2 + \text{constant}.$$

Thus, up to a parameter-independent term, training is a weighted factorization of the conditional bigram probabilities, with weights given by the input-token distribution.

2 Calculation of the Gradients and Hessian

Because x and y are fixed under parameter differentiation, the first differential of the error is

$$de = d(VWx) = dV Wx + V dW x.$$

It follows from (1) that

$$d\mathcal{L} = \mathbb{E} [e^\top (dV Wx + V dW x)].$$

Using the Frobenius inner product $\langle A, B \rangle_F = \text{tr}(A^\top B)$, the two terms satisfy

$$e^\top dV h = \langle eh^\top, dV \rangle_F, \quad e^\top V dW x = \langle V^\top e x^\top, dW \rangle_F.$$

Therefore, the population gradients are

$$\begin{aligned} \nabla_V \mathcal{L} &= \mathbb{E}[eh^\top] = \sum_{i=1}^C \pi_i (Vw_i - \boldsymbol{\pi}_{\cdot|i}) w_i^\top, \\ \nabla_W \mathcal{L} &= \mathbb{E}[V^\top e x^\top] = \sum_{i=1}^C \pi_i V^\top (Vw_i - \boldsymbol{\pi}_{\cdot|i}) \mathbf{e}_i^\top. \end{aligned} \quad (2)$$

In particular, the gradient with respect to the i -th embedding column is

$$\nabla_{w_i} \mathcal{L} = \pi_i V^\top (Vw_i - \boldsymbol{\pi}_{\cdot|i}).$$

Equation (2) shows that both the input probability π_i and the conditional target probabilities $\pi_{j|i}$ affect the gradients.

The model output is bilinear in V and W , so differentiating the error a second time gives

$$d^2 e = d^2(VWx) = 2 dV dW x.$$

The factor of two collects the two product-rule contributions. Hence the complete Hessian quadratic form is

$$d^2 \mathcal{L} = \mathbb{E} \left[\|dV Wx + V dW x\|_2^2 + 2e^\top dV dW x \right]. \quad (3)$$

Equivalently,

$$\begin{aligned} d^2 \mathcal{L} = \mathbb{E} \left[\|dV Wx\|_2^2 + \|V dW x\|_2^2 \right. \\ \left. + 2(dV Wx)^\top V dW x + 2e^\top dV dW x \right]. \end{aligned}$$

The first two terms define the two within-layer, or diagonal, Hessian blocks. The final two terms define the cross-layer blocks. The prediction error appears only in the cross-layer contribution because the model is linear in each parameter matrix separately but bilinear in the pair.

3 Hessian Block Structure and Effect of Data Imbalance

We now state the Hessian block structure and its distributional dependence. The detailed blockwise calculations are deferred to Appendix A.

Embedding-layer diagonal block. The embedding matrix is partitioned by columns along the input-vocabulary direction,

$$W = [w_1 \ \cdots \ w_C],$$

with one parameter block $w_i \in \mathbb{R}^d$ for each input token. Under the corresponding column-wise vectorization, the embedding-layer Hessian is

$$\begin{aligned} H_{WW} &= \text{BlockDiag}(\pi_1 V^\top V, \dots, \pi_C V^\top V) = D_\pi \otimes V^\top V, \\ H_{WW}^{(i)} &= \pi_i V^\top V, \quad H_{WW}[dW] = V^\top V dW D_\pi. \end{aligned} \quad (4)$$

At fixed V , all embedding blocks have the common shape $V^\top V$; their block-to-block heterogeneity is determined entirely by the input probabilities π_i . Neither the conditional target distribution nor the output marginal appears in (4). Equivalently, its input-vocabulary block structure is

$$H_{WW} = \begin{bmatrix} \pi_1 V^\top V & 0 & \cdots & 0 \\ 0 & \pi_2 V^\top V & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \pi_C V^\top V \end{bmatrix}.$$

LM-head diagonal block. The LM head is partitioned by rows along the output-vocabulary direction,

$$V = \begin{bmatrix} v_1^\top \\ \vdots \\ v_C^\top \end{bmatrix},$$

with one parameter block $v_a \in \mathbb{R}^d$ for each output token. The LM-head Hessian satisfies

$$\begin{aligned} H_{VV} &= \text{BlockDiag}(W D_\pi W^\top, \dots, W D_\pi W^\top) = I_C \otimes (W D_\pi W^\top), \\ H_{VV}^{(1)} &= \dots = H_{VV}^{(C)} = W D_\pi W^\top, \\ H_{VV}[dV] &= dV W D_\pi W^\top. \end{aligned} \quad (5)$$

Thus all output-class blocks are identical for arbitrary input and output distributions. Their common block is $WD_\pi W^\top = \sum_i \pi_i w_i w_i^\top$. Therefore, nonuniform input probabilities can unequally weight the embedding directions and induce or amplify spectral anisotropy within every common block, even though they cannot create block-to-block heterogeneity across output tokens. The target distribution does not enter (5). Equivalently, its output-vocabulary block structure is

$$H_{VV} = \begin{bmatrix} WD_\pi W^\top & 0 & \cdots & 0 \\ 0 & WD_\pi W^\top & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & WD_\pi W^\top \end{bmatrix}.$$

Cross-layer blocks. Let $v_a^\top$ be the a -th row of V , and define the conditional mean error

$$\bar{e}_{i,a} = v_a^\top w_i - \pi_{a|i}.$$

With the convention that the Hessian quadratic form contains $2 dv_a^\top H_{v_a, w_i} dw_i$, the cross-layer block is

$$H_{v_a, w_i} = \pi_i \left(w_i v_a^\top + \bar{e}_{i,a} I_d \right). \quad (6)$$

Unlike the two diagonal Hessian blocks, the cross-layer block depends on the conditional target distribution through $\bar{e}_{i,a}$. When $Vw_i = \pi_{\cdot|i}$, this residual-dependent term vanishes, while the Gauss–Newton coupling $\pi_i w_i v_a^\top$ generally remains.

Distributional dependence of the Hessian blocks

At fixed network parameters W and V :

- The heterogeneity of the embedding-layer blocks depends directly only on the input-token probabilities π_i . It is independent of $\pi_{j|i}$ and the output marginal.
- All LM-head blocks are identical and equal to $WD_\pi W^\top$. Input imbalance cannot create block-to-block heterogeneity across output tokens, but its unequal weighting of embedding directions can induce or amplify spectral heterogeneity within every common block.
- The conditional target distribution affects Hessian curvature only through the residual-dependent part of the cross-layer blocks.

A Detailed Hessian Block Calculations

This appendix derives the block formulas stated in Section 3. We start from the complete Hessian quadratic form in (3).

Embedding-layer diagonal block. Setting $dV = 0$ in (3) gives

$$d_W^2 \mathcal{L} = \mathbb{E}[\|V dW x\|_2^2].$$

Since $x = \mathbf{e}_i$, write $dw_i = dW \mathbf{e}_i$. Averaging over the input distribution gives

$$d_W^2 \mathcal{L} = \sum_{i=1}^C \pi_i dw_i^\top V^\top V dw_i.$$

There are no terms coupling dw_i and dw_k for $i \neq k$, which yields (4). Consequently, for any homogeneous matrix norm and any $\pi_k > 0$,

$$\frac{\|H_{WW}^{(i)}\|}{\|H_{WW}^{(k)}\|} = \frac{\pi_i}{\pi_k}.$$

If $\pi_i = 0$, then $H_{WW}^{(i)} = 0$, so the corresponding embedding is unidentified by the observed bigram distribution. If every $\pi_i > 0$ and V has full column rank, then

$$\kappa(H_{WW}) = \frac{\pi_{\max}}{\pi_{\min}} \kappa(V^\top V) = \frac{\pi_{\max}}{\pi_{\min}} \kappa(V)^2.$$

Thus input imbalance and the geometry of the LM head contribute multiplicatively to ill-conditioning in the embedding layer.

LM-head diagonal block. Setting $dW = 0$ in (3) gives

$$d_V^2 \mathcal{L} = \mathbb{E}[\|dV W x\|_2^2].$$

Writing the a -th row of dV as $dv_a^\top$, we obtain

$$d_V^2 \mathcal{L} = \text{tr}(dV^\top dV W D_\pi W^\top) = \sum_{a=1}^C dv_a^\top (W D_\pi W^\top) dv_a,$$

which yields (5). The target independence follows directly from

$$d_V^2 \left[\frac{1}{2} \|V w_i - y\|_2^2 \right] = \|dV w_i\|_2^2,$$

which does not depend on y .

Cross-layer blocks. The cross-layer contribution in (3) is

$$\begin{aligned} d_{VW}^2 \mathcal{L} = 2 \sum_{i=1}^C \pi_i & \left[(dV w_i)^\top V dw_i \right. \\ & \left. + (V w_i - \boldsymbol{\pi}_{\cdot|i})^\top dV dw_i \right]. \end{aligned}$$

The bilinear term between dv_a and dw_i is

$$2\pi_i dv_a^\top (w_i v_a^\top + \bar{e}_{i,a} I_d) dw_i.$$

Under the block convention stated before (6), this gives the cross-layer result in (6).